

AI시대에 맞는 보안으로 AI 혁신 가속화



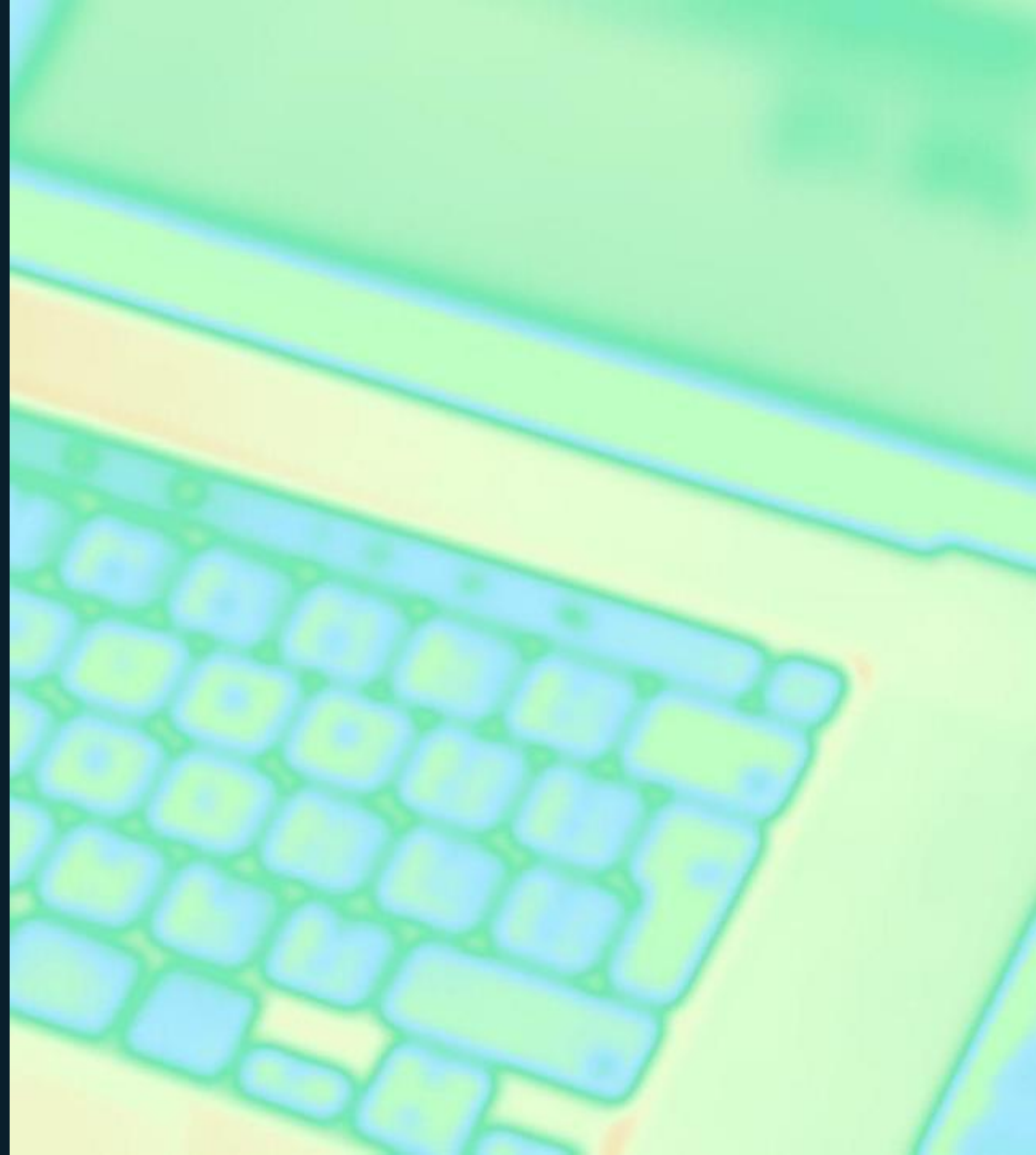
프롬프트 제출



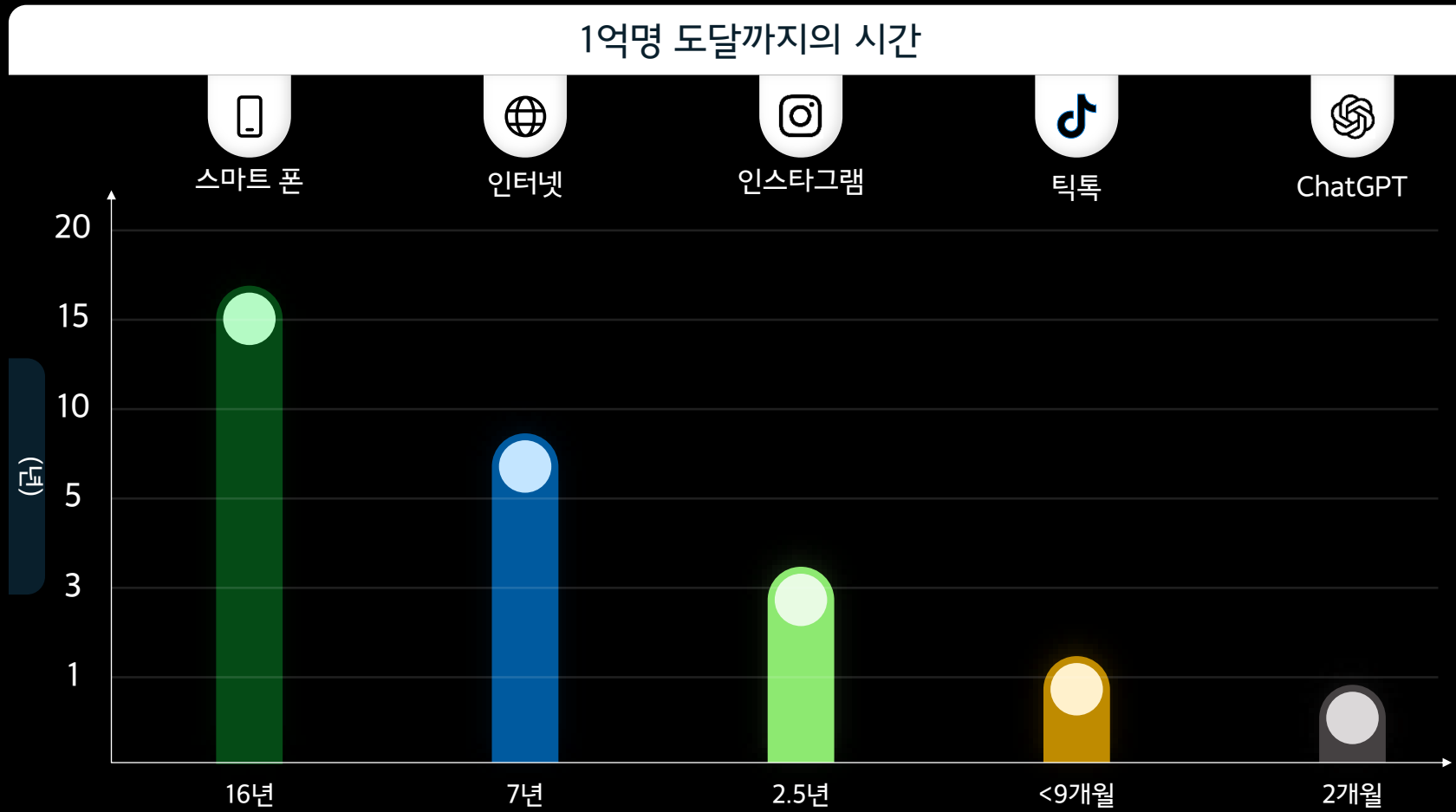
마이크로소프트 아시아 보안 GTM 박상준

보안을 위한 AI

AI for Security



생성형 AI 기술의 급성장



아울러



창의력 발휘



생산성 향상



기술 향상

AI는 패러다임의 전환을 가능하게 합니다.

보안팀의 전문성과 효율성을 폭발적으로 극대화하여 머신의 속도와 규모를 방어합니다.

대응적(사후적)



Proactive (선행적)

Siloed (사일로)



통합(통합)

정적(정적)



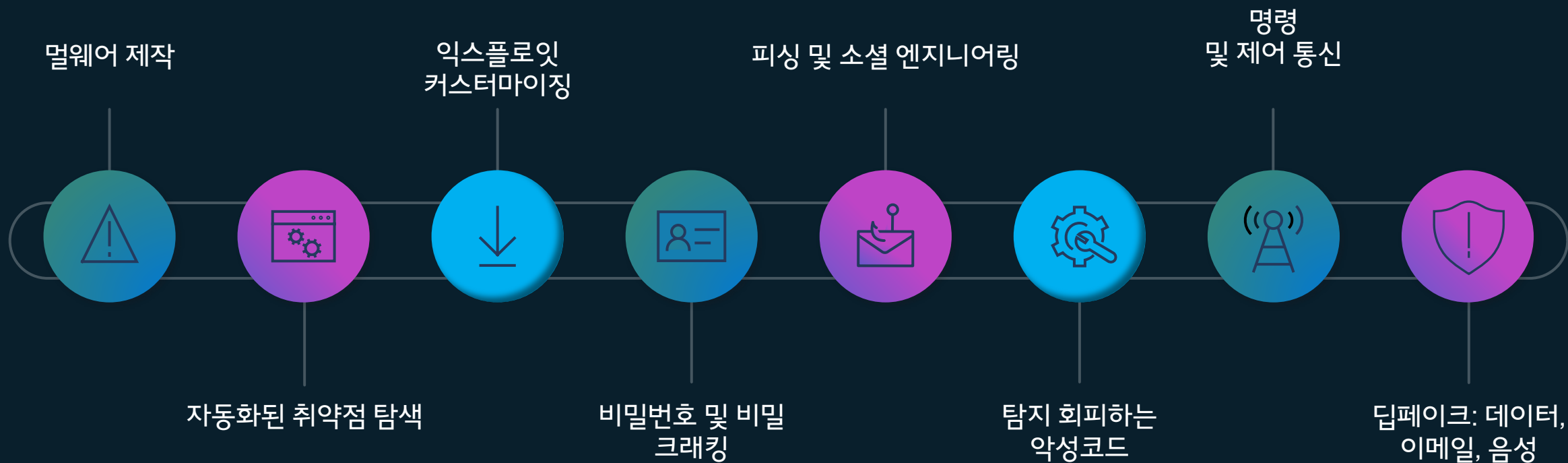
동적(능동적)

수동(수동적)



자동(자동화된)

공격자도 AI를 적극 활용합니다.



위협은 줄어들지 않고 있습니다.
공격자들은 AI를 사용하여 더 빠르고 스마트하게 움직입니다.



67%

모든 피싱 공격 중 어떤 형태로든 AI를 활용한 비율 ¹



66

조직이 하루에 처리해야 하는 데이터 보안 경고 수
(2023년에 52개에서 14개 증가) ²



73%

사이버 보안 전문가 중 중요도가 높은 보안 경고를 놓
치거나, 실수로 무시하거나, 대응에 실패한 경험의
비율 ³



AI를 통해
더 많은
것을 할 수
있음

보안은 일반 LLM보다 더 많은 것을 요구합니다.



실시간 신호 처리



조사 추론

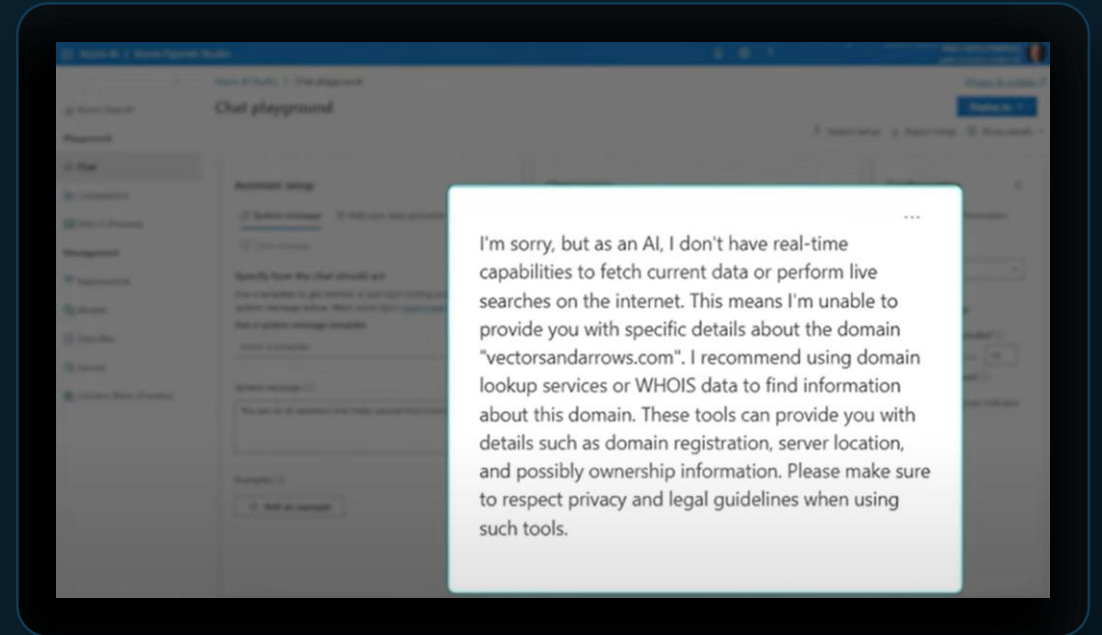


정밀도
및 증거



Persistent
데이터 컨텍스트

프롬프트: vectorsandarrows.com에 대한 평판은 어떤가요?



일반 LLM과 비교

AI 기반 사이버 보안에 대한 Microsoft의 접근 방식

AI의 능력을 향상

사이버 기술 및 프롬프트북

+

에버그린 위협 인텔리전스

+

플러그인

+

보안 특화된 오케스트레이터

+

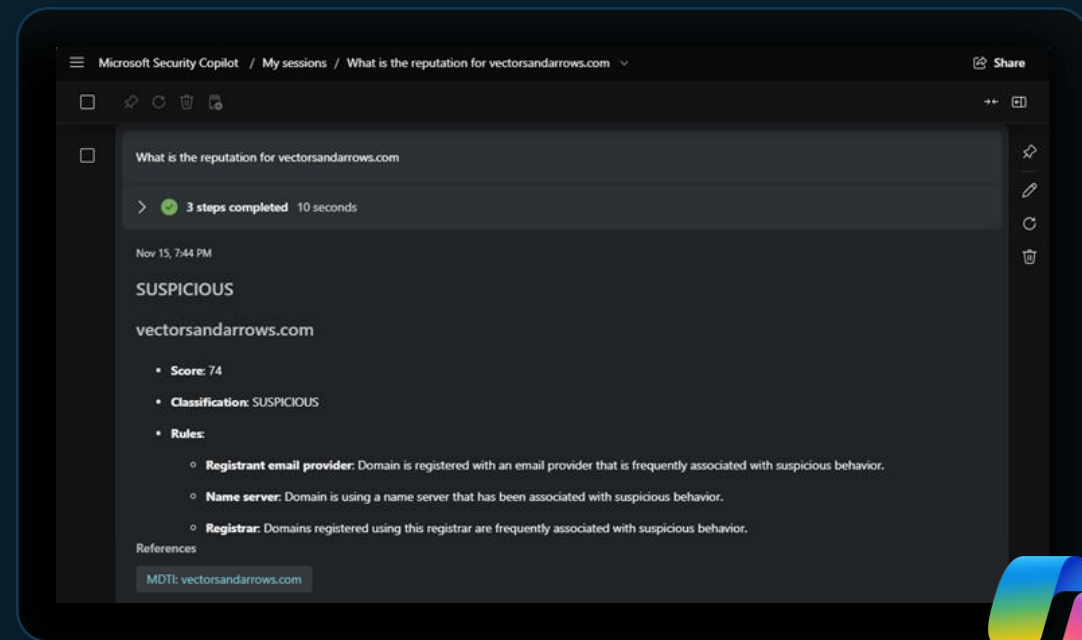
하이퍼스케일 인프라

+

고급 LLM + Microsoft 보안



프롬프트: 벡터샌드로즈닷컴에 대한 평판은 어떤가요?



Security Copilot



보안 코파일럿은 이러한 과제를 해결하기 위해 구축되었습니다.

보안 및 IT의 일상적인 운영을 위한 생성형 AI 기반 도우미로,
팀이 AI의 속도와 규모에 맞춰 보호할 수 있도록 지원합니다.



다양한 사용 사례



보안 조사
및 대응



스크립트 제작
및 리버스 엔지니어링



위험 탐색 및 태세
관리

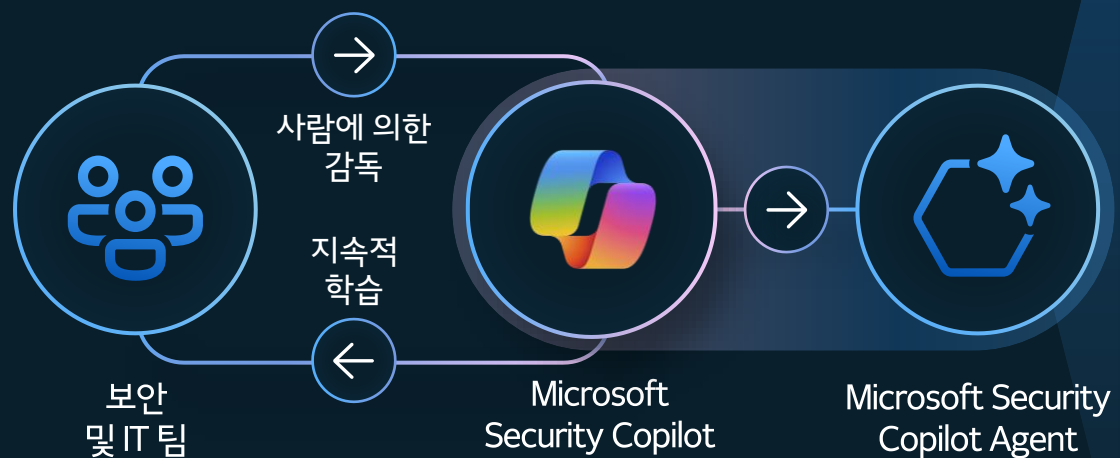


IT 문제 해결



정책
생성 및 관리

에이전트의 동적 및 자율적 자동화



Microsoft Security

 Phishing Triage

 Alert Triage in DLP and IRM

 Conditional Access Optimization

 Vulnerability Remediation

 Threat Intelligence Briefing

Partner ecosystem

 Privacy Breach Response by OneTrust

 SecOps Tooling by BlueVoyant

 Network Supervisor by Aviatrix

 Alert Triage by Tanium

 Task Optimizer by Fletch

통합 SOC 플랫폼의 코파일럿



알림 및 인시던트에
대한 인텔리전트 컨
텍스트

신속한 조사 및 대응

고급 SOC 기술 활용

Microsoft Defender

Incidents > Hands-on keyboard attack was launched from a compromised account (attack disruption)

Hands-on keyboard attack was launched from a compromised account (at...

High Active baat18@woodgrove.ms

Ransomware Critical asset Lateral Movement Attack Disruption Device Not Onboarded AlpineSkiHouse

Important! Attack disruption has automatically taken multiple response actions. For more details, go to the [Action center](#).

Attack story Alerts (44) Assets (11) Investigations (6) Evidence and Response (70) Recommended actions (21) Summary Similar incidents (0)

Alerts

- Nov 12, 2024 11:10 AM • New Potential human-operated malicious activity parkcity-win10v.parkcity.alpineskihouse.co
- Nov 12, 2024 5:43 PM • New Suspicious RDP session parkcity-win10v.parkcity.alpineskihouse.co jonaw
- Nov 12, 2024 5:43 PM • New Compromised account conducting hands-on-keyboard attack parkcity-win10v.parkcity.alpineskihouse.co jonaw
- Nov 12, 2024 5:43 PM • New Compromised account conducting hands-on-keyboard attack parkcity-win10v.parkcity.alpineskihouse.co jonaw
- Nov 12, 2024 5:45 PM • New Mimikatz credential theft tool parkcity-win10v.parkcity.alpineskihouse.co jonaw
- Nov 12, 2024 5:45 PM • New Suspicious access to LSASS service parkcity-win10v.parkcity.alpineskihouse.co jonaw
- Nov 12, 2024 5:45 PM • New Possible pass-the-hash parkcity-win10v.parkcity.alpineskihouse.co jonaw

Incident graph

Layout Group similar nodes

Logged on users (last 30 days)

Most logons jonaw Local admin

Newest logon jonaw Local admin

View all 2 logged on users

VM details

Category	Computers and Mobile	Type	Unknown
Subtype	Workstation	Discovery sources	
Domain	parkcity.alpineskiho use.co	OS	Windows 10 64-bit (Release 22H2 Build 19045.4894)
SAM name	PARKCITY-WIN10V\$	Asset group	ParkCity Alpineskihouse
Health state	Inactive	Created on	Oct 18, 2023 12:05:52 PM

Copilot

Incident summary

Nov 25, 2024 10:36 AM

The high severity incident "Hands-on keyboard attack was launched from a compromised account (attack disruption)" occurred between 2024-10-08 08:18:04 UTC and 2024-10-09 04:36:52 UTC. It was tagged as Attack Disruption, Lateral Movement, and Ransomware, triggering an automatic Attack Disruption action. This incident impacted users 'jonaw' and 'lynnr' and devices 'parkcity-win10v', 'parkcity-win10s', 'parkcity-dc', and 'parkcity-jump'.

See more

AI-generated content may be incorrect. Check it for accuracy.

Guided response

Nov 25, 2024 10:36 AM

Completed recommendations 4/11

Status: All

Triage

New

Classify this incident

Nov 25, 2024 10:36 AM

We've found similar incidents you can check to determine whether this is a 'true positive'.

Classify View evidence

AI-generated content may be incorrect. Check it for accuracy.

Containment

New

Isolate device PARKCITY-Jump.parkcity.alpineskihouse.co

Nov 25, 2024 10:36 AM

Other organizations tend to take this action for

인시던트 요약 및 조사

실행 가능한 요약 및 지침을 통해 인시던트에 대한 컨텍스트를 확보하고 더 빠르게 분류하세요.

코파일럿 이전

SOC 팀은 인시던트에 대한 컨텍스트를 수동으로 수집하는 데 상당한 시간을 소비하여 복잡한 보안 경고를 분류하고 효과적으로 대응하기가 어려웠습니다.



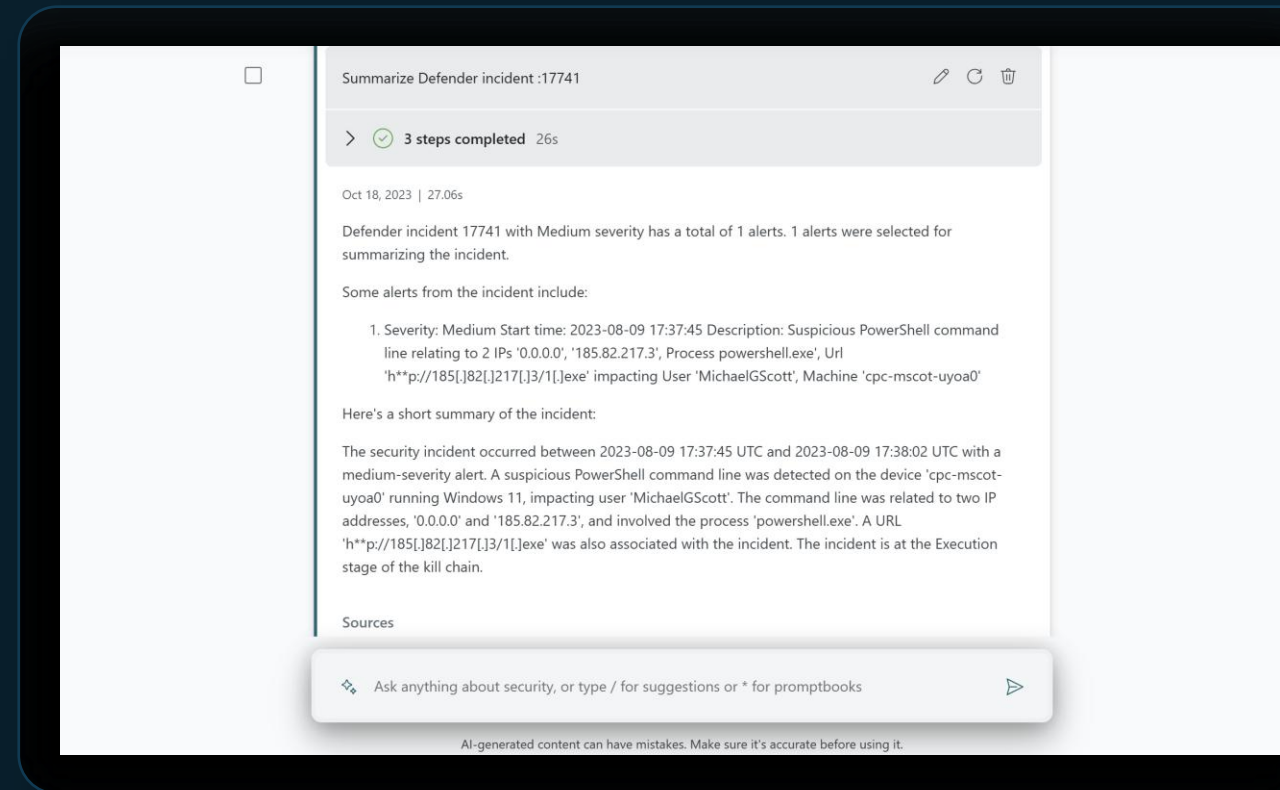
Prompt

Defender 인시던트 <xxxxx>를 글머리 기호로 요약하세요. 해당 인시던트와 관련된 엔터티에 대해 알려주세요.



보안 코파일럿

AI가 생성한 상황별 요약을 통해 주니어 SOC 분석가가 충분한 정보를 바탕으로 의사 결정을 내릴 수 있습니다.



멀웨어의 리버스 엔지니어링을 자동화하여 분석가가 공격자의 행동을 이해할 수 있도록 지원하세요.

코파일럿 이전

초급 SOC 분석가는 분석에 시간이 많이 걸리고 영향을 받는 엔터티와 연결하기 어려운 복잡한 스크립트를 이해하는 데 어려움을 겪습니다. 고급 SOC 분석가와 TI 분석가는 악성 스크립트를 수동으로 분석하는 데 너무 많은 시간을 소비하면 위협에 대한 조치가 지연됩니다.



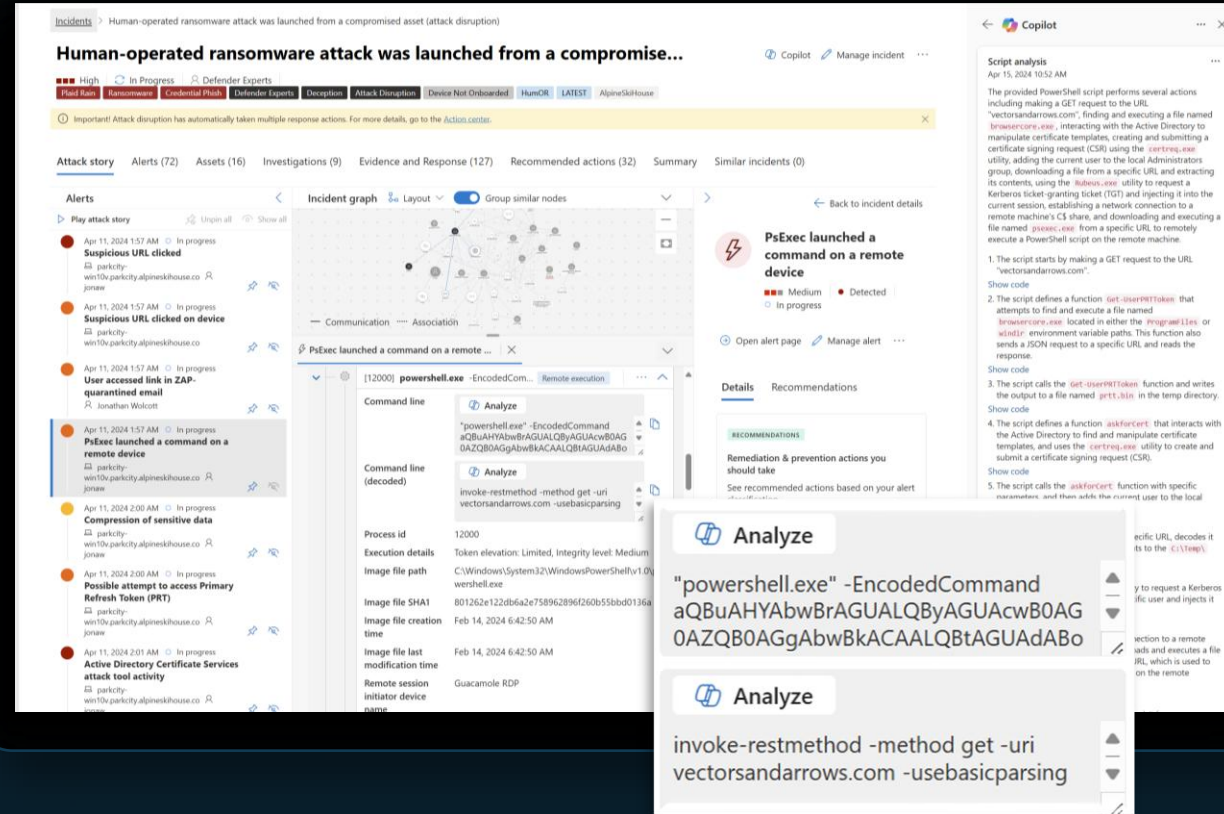
Prompt

이 스크립트를 분석하세요.
엔티티를 추출합니다.
이 URL에 대해 어떤 위협 인텔리전스를 사용할 수 있나요?
악성인가요, 그렇다면 그 이유는 무엇인가요?



보안과 파일럿

Security Copilot은 AI를 사용하여 복잡한 스크립트를 자동으로 검사하고 복잡한 스크립트를 명확하고 실행 가능한 인사이트로 변환합니다. 이를 통해 스크립트가 악성 인지 양성인지 판단하고 영향을 받는 엔터티에 지표를 연결하여 더 빠르고 정확하게 분석할 수 있습니다.



피싱 분류 에이전트 Microsoft Defender 포털에서

피싱 경고를 자율적으로 분류

정교한 평가를 수행하여 인시던트가 실제 위협인지 아니면 거짓 경보인지 매우 정확하게 판단합니다.

피드백을 기반으로 동작 진화

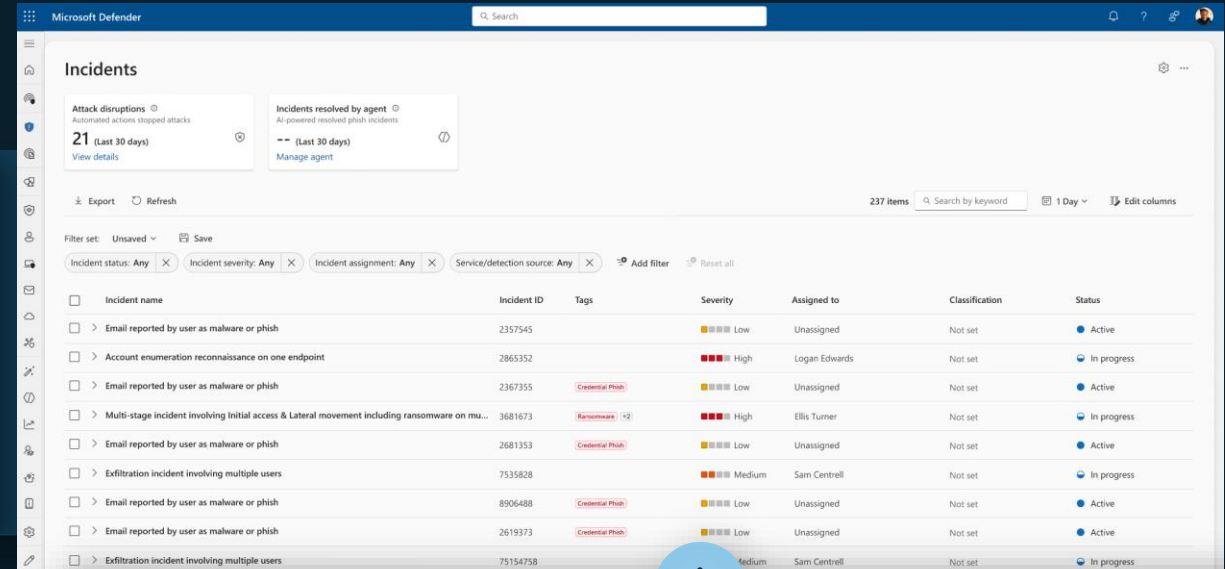
자연어로 제공되는 피드백을 통해 행동을 학습하고 진화시켜 정확도와 뉘앙스를 높이고 팀이 제어할 수 있도록 합니다.

평가에 대한 완전한 투명성 제공

평가에 대한 이유를 자연어로 제공하고 각 결론에 도달하기 위해 취한 단계를 시각적으로 매핑합니다. 팀은 필요에 따라 결정을 조정할 수 있습니다.

너무 많음

간소화



Microsoft Defender Incidents

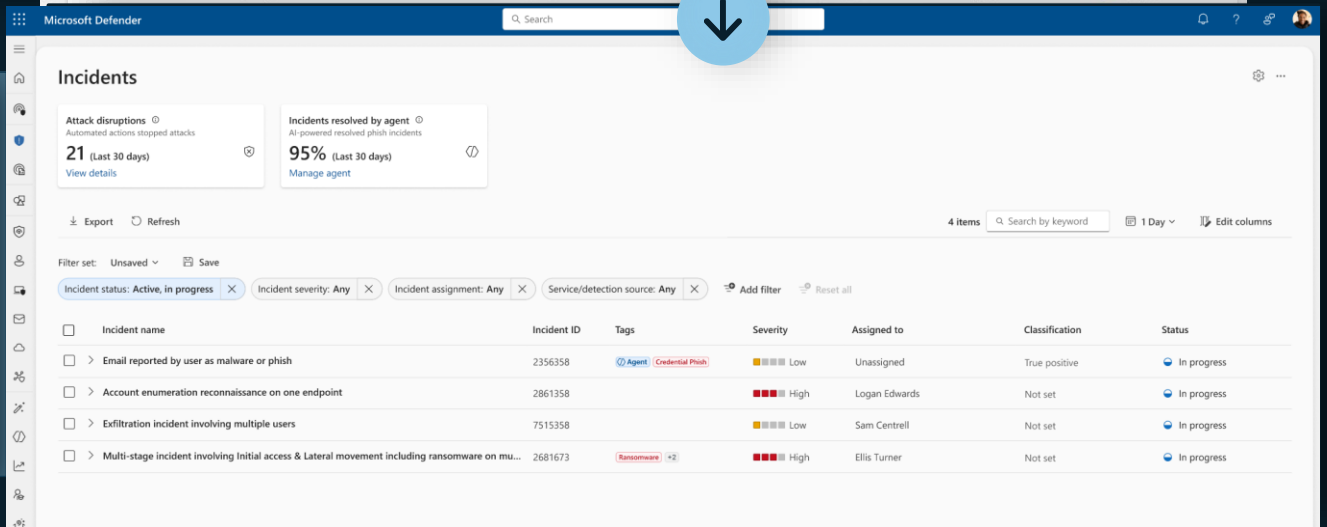
Attack disruptions: 21 (Last 30 days)
Incidents resolved by agent: -- (Last 30 days)

237 items

Filter set: Unsaved

Incident status: Any Incident severity: Any Incident assignment: Any Service/detection source: Any

Incident name	Incident ID	Tags	Severity	Assigned to	Classification	Status
Email reported by user as malware or phish	2357545		Low	Unassigned	Not set	Active
Account enumeration reconnaissance on one endpoint	2865352		High	Logan Edwards	Not set	In progress
Email reported by user as malware or phish	2367355	Credential Phish	Low	Unassigned	Not set	Active
Multi-stage incident involving Initial access & Lateral movement including ransomware on mu...	3681673	Ransomware	High	Ellis Turner	Not set	In progress
Email reported by user as malware or phish	2681353	Credential Phish	Low	Unassigned	Not set	Active
Exfiltration incident involving multiple users	7535828		Medium	Sam Centrell	Not set	In progress
Email reported by user as malware or phish	8906488	Credential Phish	Low	Unassigned	Not set	Active
Email reported by user as malware or phish	2619373	Credential Phish	Low	Unassigned	Not set	Active
Exfiltration incident involving multiple users	75154758		Medium	Sam Centrell	Not set	In progress



Microsoft Defender Incidents

Attack disruptions: 21 (Last 30 days)
Incidents resolved by agent: 95% (Last 30 days)

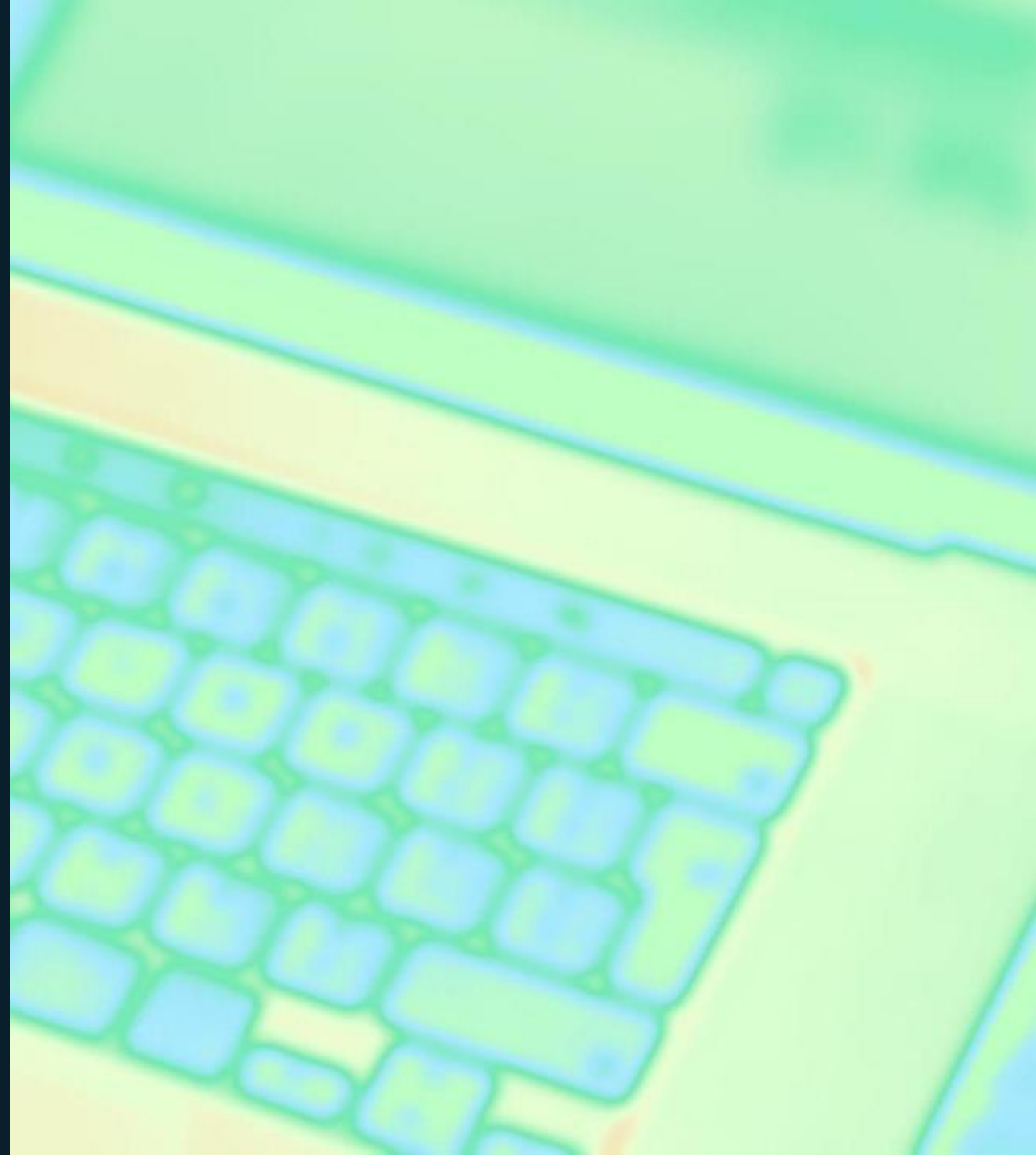
4 items

Filter set: Unsaved

Incident status: Active, in progress Incident severity: Any Incident assignment: Any Service/detection source: Any

Incident name	Incident ID	Tags	Severity	Assigned to	Classification	Status
Email reported by user as malware or phish	2356358	Agent, Credential Phish	Low	Unassigned	True positive	In progress
Account enumeration reconnaissance on one endpoint	2861358		High	Logan Edwards	Not set	In progress
Exfiltration incident involving multiple users	7515358		Low	Sam Centrell	Not set	In progress
Multi-stage incident involving Initial access & Lateral movement including ransomware on mu...	2681673	Ransomware	High	Ellis Turner	Not set	In progress

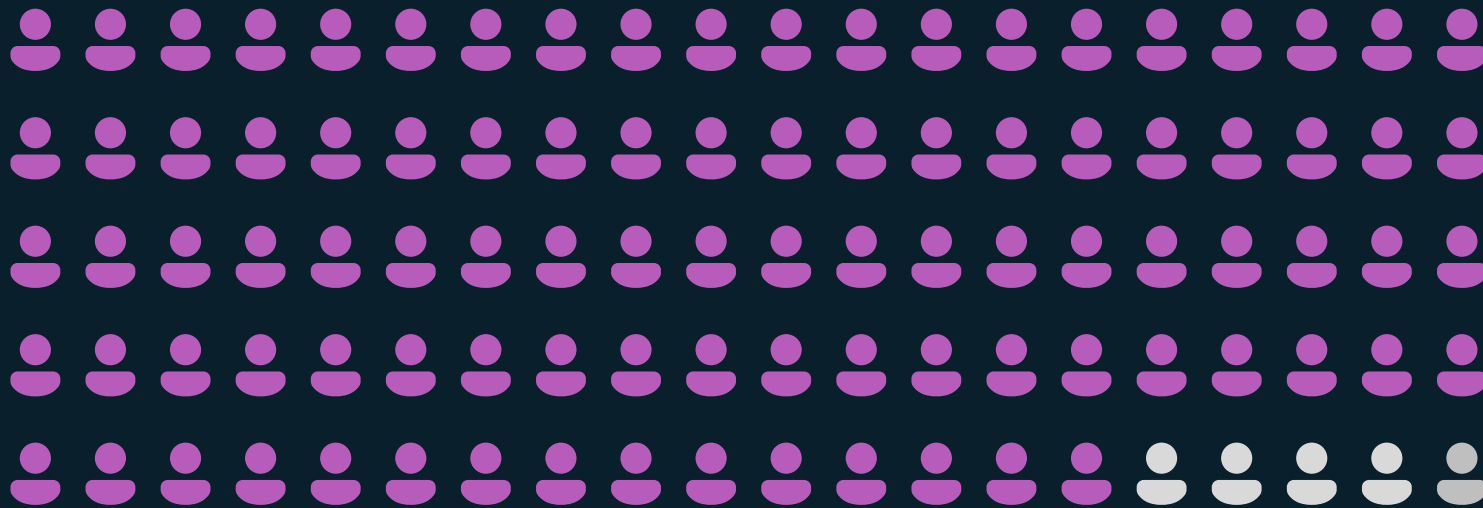
AI를 위한 보안





2026년까지 모든 애플리
케이션 중 Gen AI 포함된
비율¹ ...

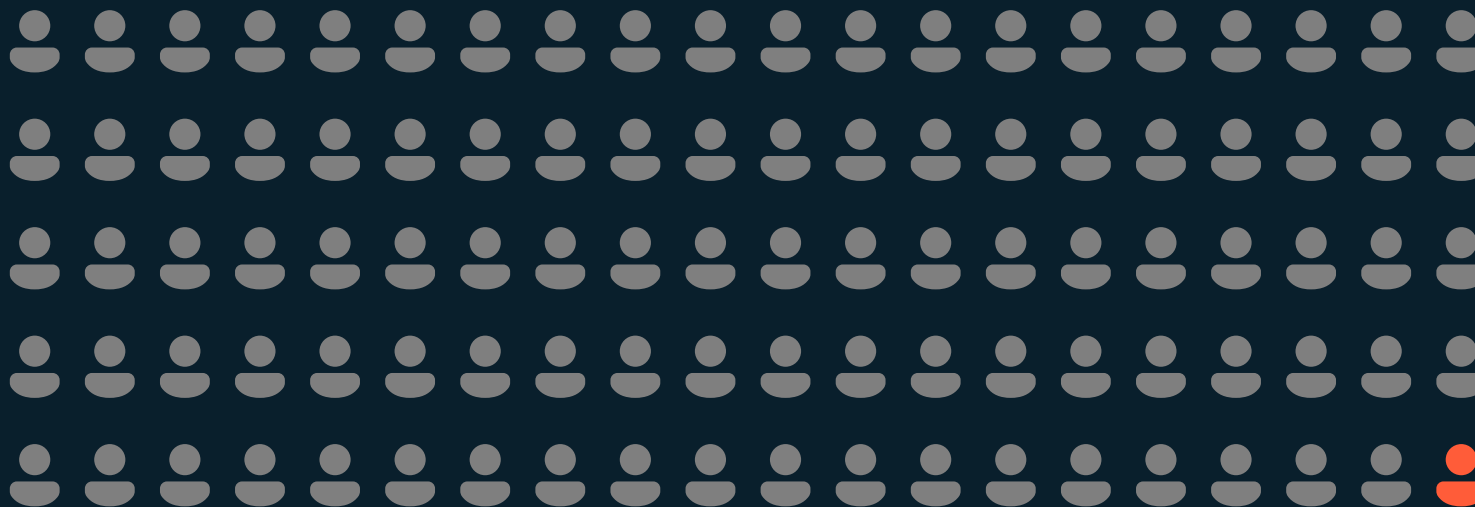
95% 이미 AI 전략을 구현 중이거나
개발 중인 기업의 비율¹



¹ Microsoft Research

1%

대규모 AI 위험에 대비하고 있는
보안 리더의 비율¹



¹ Gartner 2023

Gen AI에 대한 공격이 늘어나고 있습니다.

새로운 GenAI
공격 표면



기존
위협 벡터



Microsoft Purview

데이터 자산의 보안 및 관리

차세대 AI 생산성을 지원하는 Purview

Copilot (AI) 사용을 보호하고 관리하기 위한 도구



과도한 공유 우려
대응



데이터 손실 및
내부자 위험에 대한
대응



AI 사용에 대한 규정
준수 및 정책에 대한
거버넌스

보안

거버넌스

Home

Solutions

Learn

Settings

DSPM for AI

Overview

Recommended actions

Reports

Data assessments

Policies

Activity explorer

Solutions

Related

Information Protection

Data Loss Prevention

Insider Risk Management

Data Security Posture Management for AI

Discover and secure all AI activity in Microsoft Copilot and other AI apps. Keep your data safe and stay on track with industry regulations. [Learn more about DSPM for AI](#)

Recommendations

[View all recommendations](#)

Data security

...

Protect your data from potential oversharing risks

Use data assessments to identify potential oversharing risks in your organization. They also provide fixes to limit access to sensitive data.

View details

Sensitivity labels on data of top 100 sites

Labeled	16.6K
Not labeled	12.5K

- No sensitive information types detected
- Sensitive information types detected
- Data not scanned

Data security

...

10% of users have been detected with risky AI usage (preview)

In the last 30 days, risky AI usage has been detected from 1,000 users in your organization. Extend your insights to calculate user risk by detecting risky prompts and responses in Copilot and other AI apps.

Get started

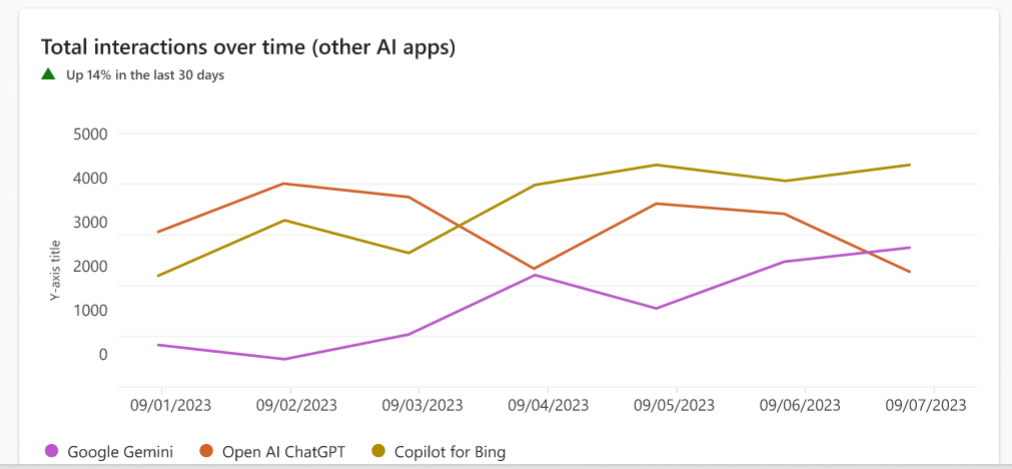
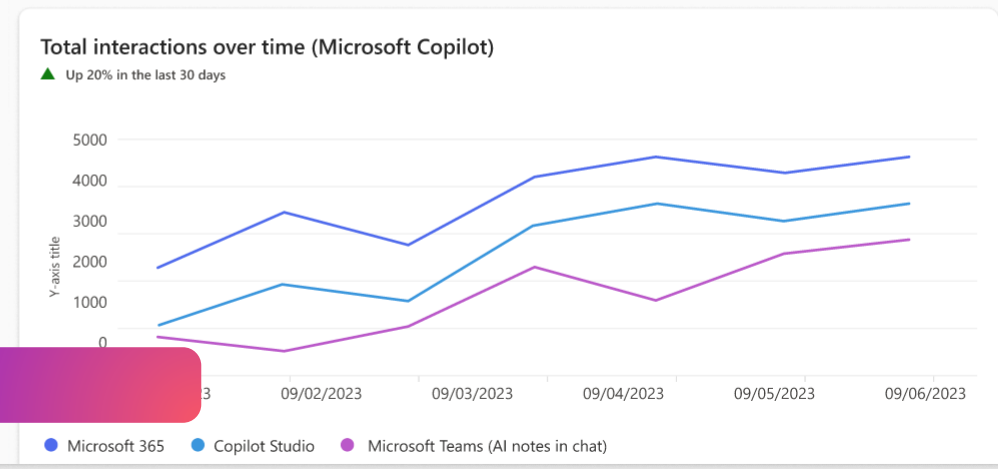
Users with risky AI usage

Last 30 days

Received sensitive prompts in Copilot	594 / 100K
Entered risky prompts in Copilot	456 / 100K

Reports

[View all reports](#)



Discovery

Project Obsidian FAQ

What is project obsidian?

- Project obsidian is a research initiative that enables scalable distributed computing.
- The project is led by a team of experts from the National Science Foundation.
- The project's main goal is to design and implement a trustworthy computation among multiple intermediaries.

What are the benefits of project obsidian?

- Project obsidian can potentially enable a scalable computation, such as machine learning.
- Project obsidian can also enhance the privacy of computation using advanced techniques such as homomorphic encryption, secure multiparty computation.
- Project obsidian can also improve the performance of computation using novel mechanisms such as sharding, consensus, and distributed storage.

How can I get involved in project obsidian?

- If you are interested in learning more about the project, you can visit the project's website at <https://projectobsidian.org>, where you can find information related to the project.
- If you are a researcher or a developer who wants to contribute to the project, you can join the project's GitHub repository at <https://github.com/projectobsidian>, where you can access the project's code, documentation, and issues.
- If you are a user or a stakeholder who wants to provide feedback or suggestions to the project, you can contact the project's team at projectobsidian@domain.com, where you can share your ideas, questions, and comments.

Search the menus

- Review selection in Editor
- Cut
- Copy
- Paste Options:
 - Expand/Collapse
- Font...
- Paragraph...
- Search "Project Obsidia..."
- Synonyms
- Rewrite with Copilot
- Read Aloud
- Translate
- Link
- New Comment

**Microsoft Purview Data Loss Prevention**
Your organization has blocked pasting protected content to unprotected locations.

You tried to paste protected content, which is prohibited by your organization.

OK

+ New chat

Recent

AI's Well-Being

Hello again

Tell me what's on your mind, or pick a suggestion.

Understand

- how something works
- rules of a sport
- what's behind a trend

Create

- game night planning
- sales pitch
- social post

Explore

- date night planning
- presentation topic ideas
- delicious noodles



Your conversations are processed by human reviewers to improve the technologies powering Bard. Don't enter anything you wouldn't want reviewed or used.

[How it works](#) [Dismiss](#)



Enter a prompt here



Bard may display inaccurate info, including about people, so double-check its responses. [Your privacy & Bard](#)

Protection

Washington, USA
[From your IP address](#) • [Update location](#)

Vertical sidebar with various application icons: search, calendar, mail, etc.

Home

Solutions

Learn

Settings

Insider Risk Management

...

Overview

Recommended actions

Alerts

Cases

Policies

Users

Reports

Forensic evidence

Adaptive protection (preview)

Notice Templates

Audit log

Solutions

Related

Communication compliance

Data Loss Prevention

Information barriers

DSPM

Remediate

Insider risk management > Alerts > (31ac5f2b) Risky AI Usage

(31ac5f2b) Risky AI Usage

Assign Needs review Confirm all alerts & create case Dismiss alert

All risk factors Activity explorer User activity Forensic evidence

Filters: Show: All scored activity for this user Risk category: Any Activity type: Any Reset all

Sort by: Date occurred

AI Usage: Prompt attacks entered in Copilot

November 1, 2023 (UTC) | Risk score: 55/100

1 events: Policy: Obsidian Merger

AI usage: Sensitive response received from M365 Copilot

November 2, 2024 (UTC) | Risk score: 85/100

10 events: Sensitive response received from Copilot

10 events: Response from sensitive site, including: Obsidian, Credit Card Number, U.S Social Security Number (SSN)

10 events: Responses that have labels applied including: General

5 events: Accessed files containing trainable classifiers including: M&A Files

(2) SEQUENCE: Label on file downgraded or removed then file accessed by Copilot

November 16 - November 20, 2024 (UTC) | Risk score: 95/100

2 events: Sequence: File label downgraded, then file accessed by Copilot

2 events: Files that have labels applied, including: Highly Confidential

2 events: Files containing sensitive info, including: Obsidian, Credit Card Number, U.S Social Security Number (SSN)

Exfiltration: File copied to removable media

November 18, 2024 (UTC) | Risk score: 75/100

5 events: File copied to removable media.

1 events: Containing sensitive info, including: Obsidian, Credit Card Number, U.S Social Security Number (SSN)

5 events: file containing labels applied, including: General

User activity scatter plot

6 Months 3 Mo

HR event: Resignation date set

(4) SEQUENCE: Label on file downgraded or removed then file accessed by Copilot

November 16 - November 20, 2024 (UTC) | Risk score: 95/100

2 events: Sequence: File label downgraded, then file accessed by Copilot

2 events: Files that have labels applied, including: Highly Confidential

2 events: Files containing sensitive info, including: Obsidian, Credit Card Number, U.S Social Security Number (SSN)

AI usage: Sensitive response received from M365 Copilot

November 18, 2024 (UTC) | Risk score: 75/100

10 events: Sensitive response received from Copilot

10 events: Response from sensitive site, including: Obsidian

10 events: Responses that have labels applied including: General

5 events: Accessed files containing trainable classifiers including: M&A Files

Obfuscation: Labels of sensitive files removed on SharePoint

November 18, 2024 (UTC) | Risk score: 55/100

1 events: 1 Labels of SharePoint files labels removed

1 events: file that have labels applied including: Highly Confidential

1 events: Containing sensitive info, including: Obsidian, Credit Card Number, U.S Social Security Number (SSN)

Nov 1, 2024

Dec 1, 2024

Jan 1, 2025

Feb 1, 2025

Mar 1, 2025

Apr 1, 2025

Access

Clean up

Collection

Exfiltration

Infiltration

Obfuscation

Risky AI usage

Sequence

Cumulative exfiltration

보안은 팀 스포츠입니다.

마이크로소프트는 보안에 올인하고 있습니다.





감사합니다